

Отчёт по лабораторной работе 03

Простейший вариант

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1 Цель работы

Изучение агентного моделирования на примере модели Д. Лавлока «Мир Маргариток», демонстрирующей механизм гомеостаза и саморегуляции в биологической системе через взаимодействие организмов с окружающей средой. # Задание

Реализовать агентную модель популяции двух типов маргариток (черные и белые) на клеточном пространстве

Реализовать механизмы роста, размножения и гибели агентов в зависимости от температуры

Реализовать обратную связь между маргаритками и температурой через альбедо

Провести вычислительные эксперименты и проанализировать динамику глобальной температуры

Визуализировать пространственное распределение маргариток и температуры

2 Теоретическое введение

Модель «Мир Маргариток» (Daisyworld) была предложена Джеймсом Лавлоком для демонстрации гипотезы Геи — идеи о том, что биота способна регулировать планетарную среду.

Основные принципы модели:

Планета заселена двумя типами маргариток: черными (низкое альбедо) и белыми (высокое альбедо)

Температура поверхности зависит от альбедо: черные маргаритки нагревают планету, белые — охлаждают

Маргаритки размножаются при оптимальной температуре и погибают при экстремальных температурах

Возникает отрицательная обратная связь, стабилизирующая глобальную температуру

Математическая основа:

Локальная температура: $T = (S \cdot (1 - \alpha) / \sigma)^{0.25} - 273.15$

Вероятность размножения: $P = 1 - |T - T_{opt}| / T_{opt}$

Глобальная температура как среднее по всем клеткам

3 Выполнение лабораторной работы

3.0.1 1. Создание нового релиза

Создаем новый релиз v.3.0.0

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ git flow release start
3.0.0
Switched to a new branch 'release/3.0.0'

Summary of actions:
- A new branch 'release/3.0.0' was created, based on 'develop'
- You are now on branch 'release/3.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

git flow release finish '3.0.0'
```

Рисунок 3.1: Создание нового релиза v.3.0.0

Обновляем информацию по релизу v.3.0.0

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ standard-changelog
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ git add CHANGELOG.md
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ git commit -m -a "shor
e(site)": update changelog
error: pathspec 'shore(site):' did not match any file(s) known to git
error: pathspec 'update' did not match any file(s) known to git
error: pathspec 'changelog' did not match any file(s) known to git
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ git commit -m -a 'chor
e(site)': update changelog
error: pathspec 'chore(site):' did not match any file(s) known to git
error: pathspec 'update' did not match any file(s) known to git
error: pathspec 'changelog' did not match any file(s) known to git
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ git commit -m -a "chore
[release/3.0.0 5eac5db] chore(site): update changelog
1 file changed, 10 insertions(+)
create mode 100644 labs/lab03/CHANGELOG.md
```

Рисунок 3.2: Обновление информации о релизе

3.0.2 2. Настройка окружения Julia

Запускаем Julia и установку пакетов

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study==simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03$ julia

Documentation: https://docs.julialang.org

Type "?" for help, "]"? for Pkg help.

Version 1.10.5 (2024-08-27)
Official https://julialang.org/ release

julia> using Pkg

julia> Pkg.add("DrWatson")
  Updating registry at `~/\.julia/registries/General.toml`
  Resolving package versions...
  No Changes to `~/\.julia/environments/v1.10/Project.toml`
  No Changes to `~/\.julia/environments/v1.10/Manifest.toml`

julia> using DrWatson

julia> initialize_project("MyProject"; authors=["Ваше Имя"], git=true)
  Activating new project at `~/work/study/2026-1/2026-1==study--simulati
on-modeling/2026-1--study--simulation-modeling/labs/lab03/MyProject`
  Resolving package versions...
  Installing Preferences — v1.5.2
  Installing ScopedValues — v1.6.0
  Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab03/MyProject/Project.toml`
[634d3b9d] + DrWatson v2.19.1
  Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab03/MyProject/Manifest.toml`
[0b6fb165] + ChunkCodecCore v1.0.1
[4c0bbe4] + ChunkCodecLibZlib v1.0.0
[55437552] + ChunkCodecLibZstd v1.0.0
[634d3b9d] + DrWatson v2.19.1
[5789e2e9] + FileIO v1.18.0
[076d061b] + HashArrayMappedTries v0.2.0
[033835bb] + JLD2 v0.6.3

```

Рисунок 3.3: Запуск Julia

3.0.3 3. Запуск скриптов и тема работы

Устанавливаем необходимые пакеты для первого скрипта


```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. src/daisyworld.jl
ERROR: LoadError: ArgumentError: Package StatsBase not found in current path.
- Run `import Pkg; Pkg.add("StatsBase")` to install the StatsBase package.
Stacktrace:
 [1] macro expansion
   @ ./loading.jl:1772 [inlined]
 [2] macro expansion
   @ ./lock.jl:267 [inlined]
 [3] __require(into::Module, mod::Symbol)
   @ Base ./loading.jl:1753
 [4] #invoke_in_world#3
   @ ./essentials.jl:926 [inlined]
 [5] invoke_in_world
   @ ./essentials.jl:923 [inlined]
 [6] require(into::Module, mod::Symbol)
   @ Base ./loading.jl:1746
in expression starting at /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld.jl:2
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. -e 'using Pkg; Pkg.add("StatsBase")'
  Resolving package versions...
  Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/Project.toml`
    [2913bbd2] + StatsBase v0.34.10
  No Changes to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/Manifest.toml`

```

Рисунок 3.4: Установка пакетов

Запуск первого скрипта src/daisyworld.jl

```

2026-1--study--simulation-modeling/labs/lab03/project/Manifest.toml
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. src/daisyworld.jl

```

Рисунок 3.5: Запуск скрипта src/daisyworld.jl

Создаем производные форматы от скрипта

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl src/daisyworld.jl
Генерация из: src/daisyworld.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld/daisyworld.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld/daisyworld.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld/daisyworld.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld/daisyworld.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld/daisyworld.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld/daisyworld.ipynb

Готово! Все файлы созданы.

```

Рисунок 3.6: Создание производных форматов

Устанавливаем необходимые пакеты для следующего скрипта

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. -e 'using Pkg; Pkg.add("CairoMakie")'
Resolving package versions...
Installed Cairo - v1.1.1
Downloaded artifact: EarCut
Downloaded artifact: Giflib
Downloaded artifact: OpenBLASConsistentFPCSR
Downloaded artifact: isoband
Downloaded artifact: Imath
Downloaded artifact: OpenEXR
Downloaded artifact: MakieAssets
Downloaded artifact: libsixel
Downloaded artifact: CRLibm
Downloaded artifact: libwebp
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/Project.toml`
[13f3f980] + CairoMakie v0.15.9
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/Manifest.toml`
[621f4979] + AbstractFFTs v1.5.0
[35492f91] + AdaptivePredicates v1.2.0
[27a7e980] + Animations v0.4.2
[67c07d97] + Automa v1.1.0
[13072b0f] + AxisAlgorithms v1.1.0
[39de3d68] + AxisArrays v0.4.8
[18cc8868] + BaseDirs v1.3.2
[96374032] + CRLibm v1.0.2
[159f3aea] + Cairo v1.1.1

```

Рисунок 3.7: Установка пакетов

Создание, запуск следующего скрипта `scripts/daisyworld_none.jl` и создание его производных форматов

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl src/daisyworld_none.jl
Генерация из: src/daisyworld_none.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld_none.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld_none/daisyworld_none.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld_none/daisyworld_none.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld_none.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld_none/daisyworld_none.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld_none/daisyworld_none.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/src/daisyworld_none.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld_none/daisyworld_none.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld_none/daisyworld_none.ipynb

Готово! Все файлы созданы.
```

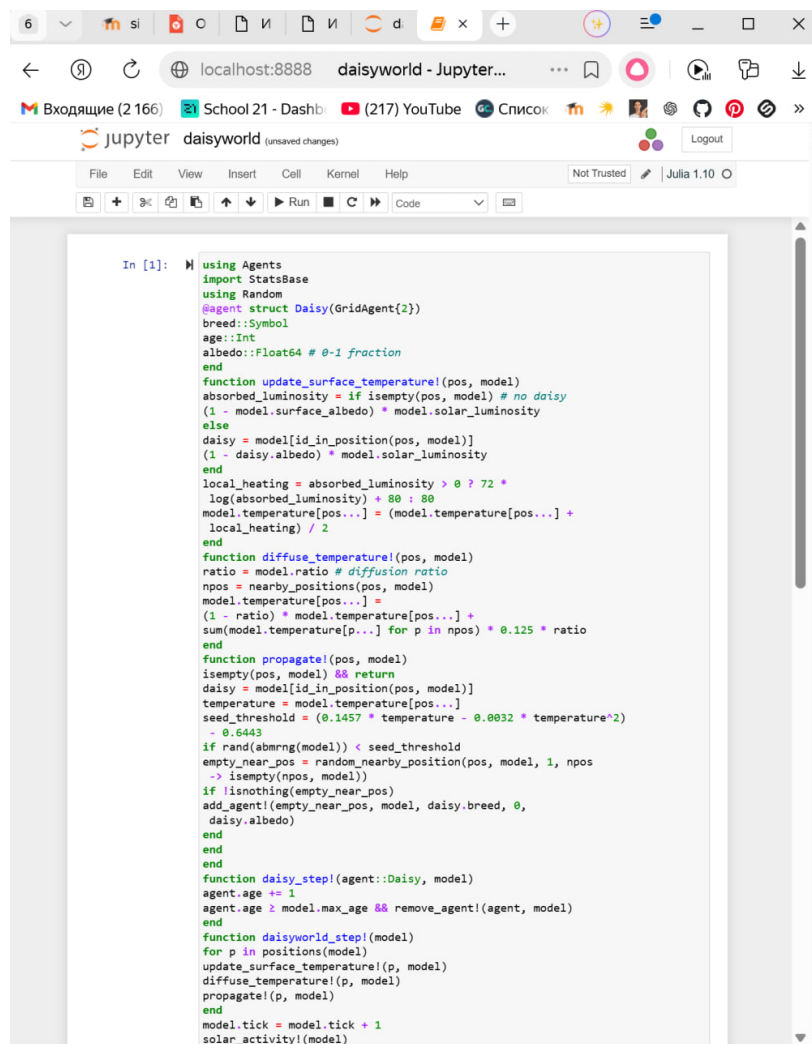
Рисунок 3.8: Запуск скрипта `scripts/daisyworld_none.jl` и создание производных форматов

Переходим в директорию `notebooks`, открываем `jupyter-notebooks` и запускаем оба проекта

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ cd notebooks/
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks$ ls
daisyworld  daisyworld_none
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks$ jupyter-notebook
[I 18:52:41.710 NotebookApp] Serving notebooks from local directory: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks
[I 18:52:41.711 NotebookApp] Jupyter Notebook 6.4.12 is running at:
[I 18:52:41.711 NotebookApp] http://localhost:8888/?token=ef19fc69a4dfa74f1b54246644f7f1c29a40eda29749ad46
[I 18:52:41.711 NotebookApp] or http://127.0.0.1:8888/?token=ef19fc69a4dfa74f1b54246644f7f1c29a40eda29749ad46
[I 18:52:41.711 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 18:52:42.959 NotebookApp]

To access the notebook, open this file in a browser:
file:///home/desti/.local/share/jupyter/runtime/nbserver-21433-o
```

Рисунок 3.9: Переход в директорию проекта и `jupyter-notebook`



```
In [1]: using Agents
import StatsBase
using Random
@agent struct Daisy(GridAgent{2})
    breed::Symbol
    age::Int
    albedo::Float64 # 0-1 fraction
end
function update_surface_temperature!(pos, model)
    absorbed_luminosity = if isempty(pos, model) # no daisy
        (1 - model.surface_albedo) * model.solar_luminosity
    else
        daisy = model[id_in_position(pos, model)]
        (1 - daisy.albedo) * model.solar_luminosity
    end
    local_heating = absorbed_luminosity > 0 ? 72 *
        log(absorbed_luminosity) + 80 : 80
    model.temperature[pos...] = (model.temperature[pos...] +
        local_heating) / 2
end
function diffuse_temperature!(pos, model)
    ratio = model.ratio # diffusion ratio
    npos = nearby_positions(pos, model)
    model.temperature[pos...] =
        (1 - ratio) * model.temperature[pos...] +
        sum(model.temperature[p...] for p in npos) * 0.125 * ratio
end
function propagate!(pos, model)
    isempty(pos, model) && return
    daisy = model[id_in_position(pos, model)]
    temperature = model.temperature[pos...]
    seed_threshold = (0.1457 * temperature - 0.0032 * temperature^2)
        - 0.6443
    if rand(abmrng(model)) < seed_threshold
        empty_near_pos = random_nearby_position(pos, model, 1, npos
        -> isempty(npos, model))
        if !isnothing(empty_near_pos)
            add_agent!(empty_near_pos, model, daisy.breed, 0,
                daisy.albedo)
        end
    end
end
function daisy_step!(agent::Daisy, model)
    agent.age += 1
    agent.age > model.max_age && remove_agent!(agent, model)
end
function daisyworld_step!(model)
    for p in positions(model)
        update_surface_temperature!(p, model)
        diffuse_temperature!(p, model)
        propagate!(p, model)
    end
    model.tick = model.tick + 1
    solar_activity!(model)
```

Рисунок 3.10: jupyter-notebook от scripts/daisyworld.jl

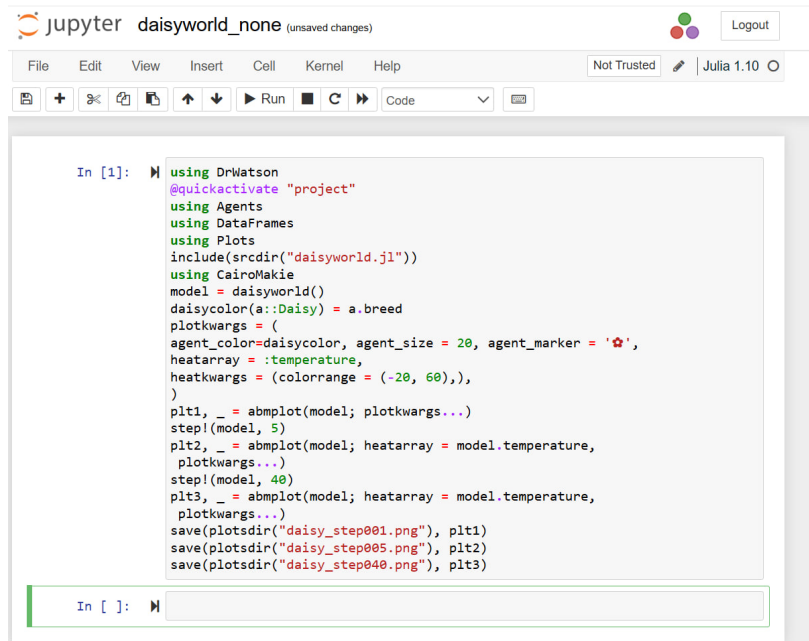


Рисунок 3.11: jupyter-notebook от scripts/daisyworld_none.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld-animate.jl.

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/project$ nano scripts/d
aisyworld-animate.jl.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/project$ julia --projec
t=. scripts/daisyworld-animate.jl.

```

Рисунок 3.12: Запуск скрипта scripts/daisyworld_animate.jl

Перехожу в директорию plots и проверяю, что нужные файлы создались

```

Project.toml  _research_data  notebooks  plots  src
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/project$ cd plots
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/project/plots$ ls
daisy_step001.png  daisy_step005.png  daisy_step040.png  simulation.mp4

```

Рисунок 3.13: Сгенерированные файлы в директории plots

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld-count.jl, генерирую производные форматы, перехожу в jupyter-notebook и запускаю проект

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--st
udy--simulation-modeling/labs/lab03/project$ nano scripts/daisyworld-count.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--st
udy--simulation-modeling/labs/lab03/project$ julia --project=. scripts/daisyworld-cou
nt.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--st
udy--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl scri
pts/daisyworld-count.jl
Генерация из: scripts/daisyworld-count.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulat
ion-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld
-count.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count/daisyworl
d-count.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/20
26-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count/daisywor
ld-count.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-
modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-cou
nt.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-count/daisywor
ld-count.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--s
tudy--simulation-modeling/labs/lab03/project/markdown/daisyworld-count/daisyworld-cou
nt.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-model
ing/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-count/daisywo
rld-count.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1-
-study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-count/daisyworld-
count.ipynb

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--st
udy--simulation-modeling/labs/lab03/project$ jupyter-notebook
[I 19:04:16.422 NotebookApp] The port 8888 is already in use, trying another port.
[I 19:04:16.423 NotebookApp] Serving notebooks from local directory: /home/desti/work
/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/l
abs/lab03/project
[I 19:04:16.423 NotebookApp] Jupyter Notebook 6.4.12 is running at:
[I 19:04:16.423 NotebookApp] http://localhost:8889/?token=6ec96a825f11108deec562d55b4
29e5a1dcdf69d9df32bbd
[I 19:04:16.423 NotebookApp] or http://127.0.0.1:8889/?token=6ec96a825f11108deec562d
55b429e5a1dcdf69d9df32bbd
[I 19:04:16.423 NotebookApp] Use Control-C to stop this server and shut down all kern
els (twice to skip confirmation).
[C 19:04:17.686 NotebookApp]

```

Рисунок 3.14: Запуск скрипта `scripts/daisyworld_count.jl` и создание производных форматов

```
In [1]: using DrWatson
        @quickactivate "project"
        using Agents
        using DataFrames
        using Plots
        include(scrdir("daisyworld.jl"))
        using CairoMakie
        black(a) = a.breed == :black
        white(a) = a.breed == :white
        adata = [(black, count), (white, count)]
        model = daisyworld(; solar_luminosity = 1.0)
        agent_df, model_df = run!(model, 1000; adata)
        figure = Figure(size = (600, 400));
        ax = figure[1, 1] = Axis{Figure}(figure, xlabel = "tick", ylabel = "daisy
        count")
        blackl = lines!(ax, agent_df[!, :time], agent_df[!, :count_black],
        color = :black)
        whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white],
        color = :orange)
        Legend{Figure}(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz
        e = 12)

Out[1]: Legend()

figure

In [2]: save(plotsdir("daisy_count.png"), figure)

In [ ]: |
```

Рисунок 3.15: jupyter-notebook от scripts/daisyworld_count.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld-luminosity.jl, генерирую производные форматы, пер>

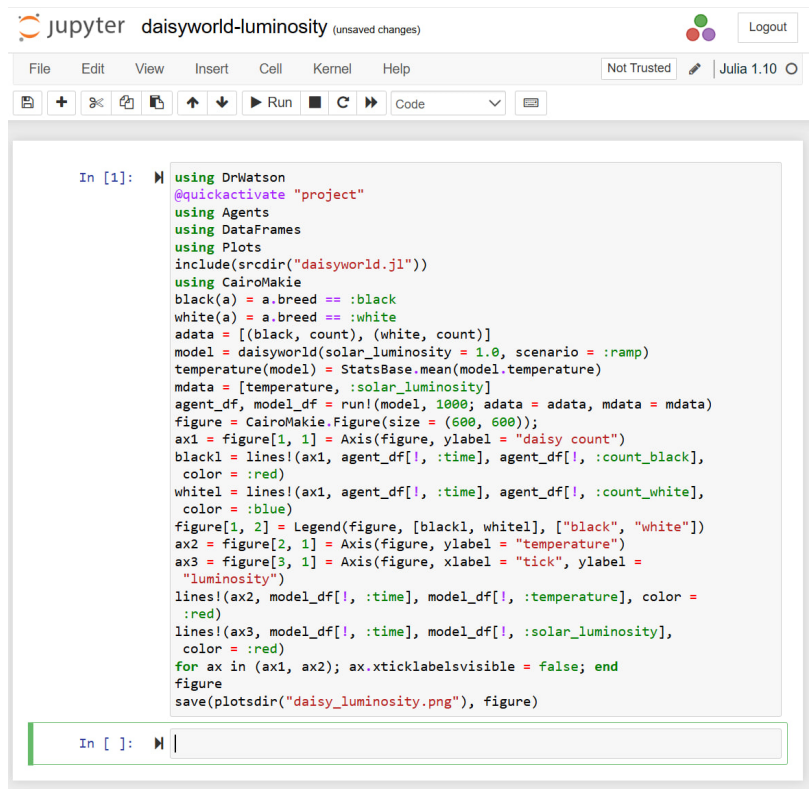
```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ nano scripts/daisyworld-luminosity.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/daisyworld-luminosity.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl scripts/daisyworld-luminosity.jl
Генерация из: scripts/daisyworld-luminosity.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity/daisyworld-luminosity.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity/daisyworld-luminosity.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-luminosity/daisyworld-luminosity.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-luminosity/daisyworld-luminosity.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-luminosity/daisyworld-luminosity.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-luminosity/daisyworld-luminosity.ipynb

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ jupyter-notebook
[I 19:11:29.563 NotebookApp] The port 8888 is already in use, trying another port.
[I 19:11:29.563 NotebookApp] The port 8889 is already in use, trying another port.
[I 19:11:29.564 NotebookApp] Serving notebooks from local directory: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-

```

Рисунок 3.16: Запуск скрипта scripts/daisyworld_luminosity.jl и создание производных форматов



```
In [1]: using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(scrdir("daisyworld.jl"))
using CairoMakie
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(solar_luminosity = 1.0, scenario = :ramp)
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]
agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)
figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
blackl = lines!(ax1, agent_df[!, :time], agent_df[!, :count_black],
color = :red)
whitel = lines!(ax1, agent_df[!, :time], agent_df[!, :count_white],
color = :blue)
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])
ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel =
"luminosity")
lines!(ax2, model_df[!, :time], model_df[!, :temperature], color =
:red)
lines!(ax3, model_df[!, :time], model_df[!, :solar_luminosity],
color = :red)
for ax in (ax1, ax2); ax.xtickslabelsvisible = false; end
figure
save(plotsdir("daisy_luminosity.png"), figure)
```

Рисунок 3.17: jupyter-notebook от scripts/daisyworld_luminosity.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld__param.jl, генерирую производные форматы>

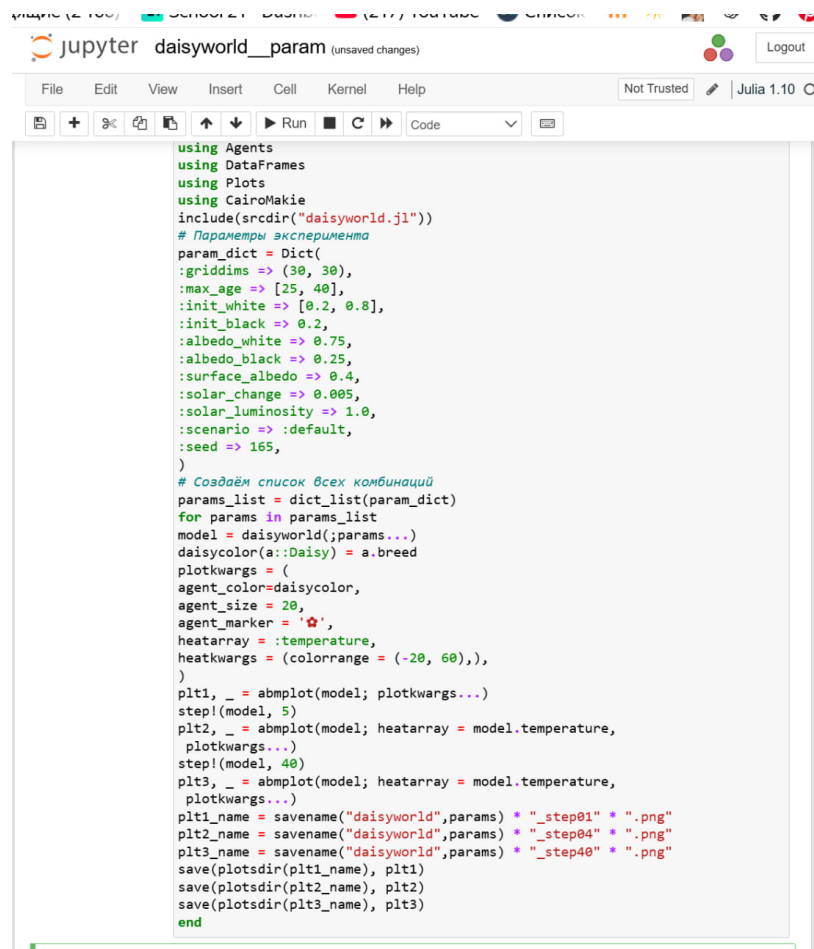
```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ nano scripts/daisyworld__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/daisyworld__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl scripts/daisyworld__param.jl
Генерация из: scripts/daisyworld__param.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param/daisyworld__param.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param/daisyworld__param.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld__param/daisyworld__param.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld__param/daisyworld__param.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld__param/daisyworld__param.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld__param/daisyworld__param.ipynb

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ jupyter-notebook
[I 19:15:49.320 NotebookApp] The port 8888 is already in use, trying another port.
[I 19:15:49.321 NotebookApp] The port 8889 is already in use, trying another port.
[I 19:15:49.321 NotebookApp] The port 8890 is already in use, trying another port.
[I 19:15:49.322 NotebookApp] Serving notebooks from local directory: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-

```

Рисунок 3.18: Запуск скрипта scripts/daisyworld__param.jl и создание производных форматов



```
using Agents
using DataFrames
using Plots
using CairoMakie
include(scrdir("daisyworld.jl"))
# Параметры эксперимента
param_dict = Dict{
  :griddims => (30, 30),
  :max_age => [25, 40],
  :init_white => [0.2, 0.8],
  :init_black => 0.2,
  :albedo_white => 0.75,
  :albedo_black => 0.25,
  :surface_albedo => 0.4,
  :solar_change => 0.005,
  :solar_luminosity => 1.0,
  :scenario => :default,
  :seed => 165,
}
# Создаём список всех комбинаций
params_list = dict_list(param_dict)
for params in params_list
  model = daisyworld(;params...)
  daisycolor(a::Daisy) = a.breed
  plotkwargs = (
    agent_color=daisycolor,
    agent_size = 20,
    agent_marker = '☆',
    heatmap = :temperature,
    heatmapkwargs = (colorrange = (-20, 60),),
  )
  plt1, _ = abmplot(model; plotkwargs...)
  step!(model, 5)
  plt2, _ = abmplot(model; heatmap = model.temperature,
    plotkwargs...)
  step!(model, 40)
  plt3, _ = abmplot(model; heatmap = model.temperature,
    plotkwargs...)
  plt1_name = savename("daisyworld", params) * "_step01" * ".png"
  plt2_name = savename("daisyworld", params) * "_step04" * ".png"
  plt3_name = savename("daisyworld", params) * "_step40" * ".png"
  save(plotsdir(plt1_name), plt1)
  save(plotsdir(plt2_name), plt2)
  save(plotsdir(plt3_name), plt3)
end
```

Рисунок 3.19: jupyter-notebook от scripts/daisyworld__param.jl

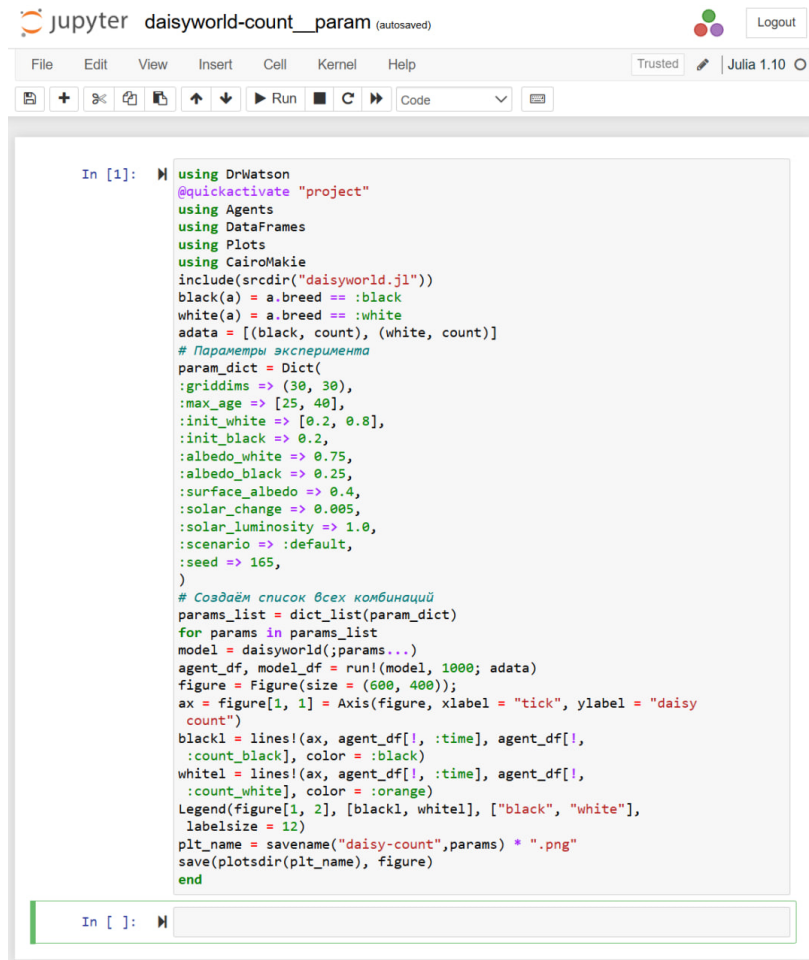
Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld-count__param.jl, генерирую производные форматы>

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ nano scripts/daisyworld-count__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/daisyworld-count__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tangle.jl scripts/daisyworld-count__param.jl
Генерация из: scripts/daisyworld-count__param.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param/daisyworld-count__param.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param/daisyworld-count__param.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-count__param/daisyworld-count__param.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-count__param/daisyworld-count__param.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-count__param/daisyworld-count__param.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-count__param/daisyworld-count__param.ipynb
Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ jupyter-notebook
[I 19:19:14.001 NotebookApp] The port 8888 is already in use, trying another port.
[I 19:19:14.001 NotebookApp] The port 8889 is already in use, trying another port.
[I 19:19:14.001 NotebookApp] The port 8890 is already in use, trying another port.

```

Рисунок 3.20: Запуск скрипта scripts/daisyworld-count__param.jl и создание производных форматов



The screenshot shows a Jupyter Notebook window titled "daisyworld-count__param (autosaved)". The interface includes a top menu bar with options like File, Edit, View, Insert, Cell, Kernel, and Help. Below the menu is a toolbar with icons for file operations and execution. The main area contains a code cell with the following Julia code:

```
In [1]: using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(scrdir("daisyworld.jl"))
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

# Параметры эксперимента
param_dict = Dict{
  :griddims => (30, 30),
  :max_age => [25, 40],
  :init_white => [0.2, 0.8],
  :init_black => 0.2,
  :albedo_white => 0.75,
  :albedo_black => 0.25,
  :surface_albedo => 0.4,
  :solar_change => 0.005,
  :solar_luminosity => 1.0,
  :scenario => :default,
  :seed => 165,
}

# Создаём список всех комбинаций
params_list = dict_list(param_dict)
for params in params_list
  model = daisyworld(;params...)
  agent_df, model_df = run!(model, 1000; adata)
  figure = Figure(size = (600, 400));
  ax = figure[1, 1] = Axis{Figure, xlabel = "tick", ylabel = "daisy
  count"}
  blackl = lines!(ax, agent_df[!, :time], agent_df[!,
    :count_black], color = :black)
  whitel = lines!(ax, agent_df[!, :time], agent_df[!,
    :count_white], color = :orange)
  Legend{Figure, 2, [blackl, whitel], ["black", "white"],
    labels = 12}
  plt_name = savename("daisy-count", params) * ".png"
  save(plotsdir(plt_name), figure)
end
```

At the bottom of the code cell, there is an input prompt "In []:" followed by a cursor.

Рисунок 3.21: jupyter-notebook от scripts/daisyworld-count__param.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт nano scripts/daisyworld-luminosity__param.jl, генерирую производные форма>

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ nano scripts/daisyworld-lumino
sity__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/dai
syworld-luminosity__param.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. scripts/tang
le.jl scripts/daisyworld-luminosity__param.jl
Генерация из: scripts/daisyworld-luminosity__param.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--si
mulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/
daisyworld-luminosity__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modelin
g/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-lumi
nosity__param/daisyworld-luminosity__param.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-lumi
nosity__param/daisyworld-luminosity__param.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simula
tion-modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/dais
yworld-luminosity__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modelin
g/2026-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-lumi
nosity__param/daisyworld-luminosity__param.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026
-1--study--simulation-modeling/labs/lab03/project/markdown/daisyworld-luminosity
__param/daisyworld-luminosity__param.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-
modeling/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworl
d-luminosity__param.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modelin
g/2026-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-lum
inosity__param/daisyworld-luminosity__param.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/20
26-1--study--simulation-modeling/labs/lab03/project/notebooks/daisyworld-luminos
ity__param/daisyworld-luminosity__param.ipynb

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ jupyter-notebook
[I 19:28:04.975 NotebookApp] The port 8888 is already in use, trying another por
t.
[I 19:28:04.976 NotebookApp] The port 8889 is already in use, trying another por
t.
[I 19:28:04.976 NotebookApp] The port 8890 is already in use, trying another por
t.

```

Рисунок 3.22: Запуск скрипта scripts/daisyworld-luminosity__param.jl и созда-
ние производных форматов

```

In [1]: using DrWatson
        @quickactivate "project"
        using Agents
        using DataFrames
        using Plots
        using CairoMakie
        include(scrdir("daisyworld.jl"))
        black(a) = a.breed == :black
        white(a) = a.breed == :white
        adata = [(black, count), (white, count)]
        # Параметры эксперимента
        param_dict = Dict{
            :griddims => (30, 30),
            :max_age => [25, 40],
            :init_white => [0.2, 0.8],
            :init_black => 0.2,
            :albedo_white => 0.75,
            :albedo_black => 0.25,
            :surface_albedo => 0.4,
            :solar_change => 0.005,
            :solar_luminosity => 1.0,
            :scenario => :ramp,
            :seed => 165,
        }
        # Создаём список всех комбинаций
        params_list = dict_list(param_dict)
        for params in params_list
            model = daisyworld(params...)
            temperature(model) = StatsBase.mean(model.temperature)
            mdata = [temperature, :solar_luminosity]
            agent_df, model_df = run!(model, 1000; adata = adata, mdata =
            mdata)
            figure = CairoMakie.Figure(size = (600, 600));
            ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
            blackl = lines!(ax1, agent_df[:, :time], agent_df[:,
            :count_black], color = :red)
            whitel = lines!(ax1, agent_df[:, :time], agent_df[:,
            :count_white], color = :blue)
            figure[1, 2] = Legend(figure, [blackl, whitel], ["black",
            "white"])
            ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
            ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel =
            "luminosity")
            lines!(ax2, model_df[:, :time], model_df[:, :temperature], color
            = :red)
            lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity],
            color = :red)
            for ax in (ax1, ax2); ax.xtickslabelsvisible = false; end
            plt_name = savename("daisy-luminosity", params) * ".png"
            save(plotsdir(plt_name), figure)
        end

```

Рисунок 3.23: jupyter-notebook от scripts/daisyworld-luminosity__param.jl

В каталоге отчёта в файл `_quarto.yml` включаем поддержку кода julia

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab03/report$ nano _quarto.y
ml

```

Рисунок 3.24: Настройка `_quarto.yml`

В файле отчёта подключаем файл описания программ и компилируем от-
чет

```

{{< include ../project/markdown/daisyworld/daisyworld.qmd >}}
{{< include ../project/markdown/daisyworld_none/daisyworld_none.qmd >}}
{{< include ../project/markdown/daisyworld-count/daisyworld-count.qmd >}}
{{< include ../project/markdown/daisyworld__param/daisyworld__param.qmd >}}
{{< include ../project/markdown/daisyworld-count__param/daisyworld-count__param.qmd >>
{{< include ../project/markdown/daisyworld-luminosity/daisyworld-luminosity.qmd >}}
{{< include ../project/markdown/daisyworld-luminosity__param/daisyworld-luminosity__param.qmd >}}

```

Рисунок 3.25: Подключение файлов

```

(base) ~/work/study/2024-1/2024-1--study--simulation-modeling/2024-1--study--simulation-modeling/lab03/reports$ quarto render simulation-modeling-lab03-report.
quarto-to-pdf --no-execute
pandoc
to: latex
output-file: simulation-modeling-lab03-report.tex
standalone: true
self-contained: true
pdf-engine: xelatex
variables:
  graphics: true
  tables: true
default-image-extension: pdf
number-sections: true
toc: true
toc-depth: 2
cite-method: biblatex
metadata
documentclass: scrreprt
classoption:
  - DIV11
  - numbersnoendperiod
papersize: a4
header-includes:
  - \usepackage{fontspec}[fontspec]

```

Рисунок 3.26: Компиляция

```

using Agents
import StatsBase
using Random

@agent struct Daisy(GridAgent{2})
  breed::Symbol
  age::Int
  albedo::Float64 # 0-1 fraction
end

function update_surface_temperature!(pos, model)
  absorbed_luminosity = if isempty(pos, model) # no daisy
    (1 - model.surface_albedo) * model.solar_luminosity
  else
    daisy = model[id_in_position(pos, model)]
    (1 - daisy.albedo) * model.solar_luminosity
  end
  local_heating = absorbed_luminosity > 0 ? 72 *
    log(absorbed_luminosity) + 80 : 80
  model.temperature[pos...] = (model.temperature[pos...] +

```



```

    local_heating) / 2
end
function diffuse_temperature!(pos, model)
    ratio = model.ratio # diffusion ratio
    npos = nearby_positions(pos, model)
    model.temperature[pos...] =
    (1 - ratio) * model.temperature[pos...] +
    sum(model.temperature[p...] for p in npos) * 0.125 * ratio
end
function propagate!(pos, model)
    isempty(pos, model) && return
    daisy = model[id_in_position(pos, model)]
    temperature = model.temperature[pos...]
    seed_threshold = (0.1457 * temperature - 0.0032 * temperature^2)
    - 0.6443
    if rand(abmrng(model)) < seed_threshold
        empty_near_pos = random_nearby_position(pos, model, 1, npos
        -> isempty(npos, model))
        if !isnothing(empty_near_pos)
            add_agent!(empty_near_pos, model, daisy.breed, 0,
            daisy.albedo)
        end
    end
end
end
end
function daisy_step!(agent::Daisy, model)
    agent.age += 1
    agent.age ≥ model.max_age && remove_agent!(agent, model)
end

```

```

function daisyworld_step!(model)
for p in positions(model)
update_surface_temperature!(p, model)
diffuse_temperature!(p, model)
propagate!(p, model)
end
model.tick = model.tick + 1
solar_activity!(model)
end
function solar_activity!(model)
if model.scenario == :ramp
if model.tick > 200 && model.tick ≤ 400
model.solar_luminosity += model.solar_change
end
if model.tick > 500 && model.tick ≤ 750
model.solar_luminosity -= model.solar_change / 2
end
elseif model.scenario == :change
model.solar_luminosity += model.solar_change
end
end
using Random
function daisyworld(;
griddims = (30, 30),
max_age = 25,
init_white = 0.2, # % cover of the world surface of white breed
init_black = 0.2, # % cover of the world surface of black breed
albedo_white = 0.75,

```

```

albedo_black = 0.25,
surface_albedo = 0.4,
solar_change = 0.005,
solar_luminosity = 1.0, # initial luminosity
scenario = :default,
seed = 165,
)
rng = MersenneTwister(seed)
space = GridSpaceSingle(griddims)
properties = (;max_age, surface_albedo, solar_luminosity,
  solar_change, scenario,
tick = 0, ratio = 0.5, temperature = zeros(griddims)
)
properties = Dict{k=>v for (k,v) in pairs(properties))
model = StandardABM(Daisy, space; properties, rng, agent_step! =
  daisy_step!, model_step! = daisyworld_step!)
grid = collect(positions(model))
num_positions = prod(griddims)
white_positions =
StatsBase.sample(grid, Int(init_white * num_positions);
  replace = false)
for wp in white_positions
add_agent!(wp, Daisy, model, :white, rand(abmrng(model),
  0:max_age), albedo_white)
end
allowed = setdiff(grid, white_positions)
black_positions =
StatsBase.sample(allowed, Int(init_black * num_positions);

```

```

    replace = false)
for bp in black_positions
add_agent!(bp, Daisy, model, :black, rand(abmrng(model),
    0:max_age), albedo_black)
end
for p in positions(model)
update_surface_temperature!(p, model)
end
return model
end

```

```

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(scrdir("daisyworld.jl"))
using CairoMakie
model = daisyworld()
daisycolor(a::Daisy) = a.breed
plotkwargs = (
agent_color=daisycolor, agent_size = 20, agent_marker = '□',
heatarray = :temperature,
heatkwargs = (colorrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature,

```

```

    plotkwargs...)
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature,
    plotkwargs...)
save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)

```

```

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(sourcedir("daisyworld.jl"))
using CairoMakie

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(; solar_luminosity = 1.0)
agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy
    count")
blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black],
    color = :black)
whitel = lines!(ax, agent_df[:, :time], agent_df[:, :count_white],
    color = :orange)
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz

```

```
= 12)
```

```
figure
```

```
save(plotsdir("daisy_count.png"), figure)
```

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(sourcedir("daisyworld.jl"))
# Параметры эксперимента
param_dict = Dict(
  :griddims => (30, 30),
  :max_age => [25, 40],
  :init_white => [0.2, 0.8],
  :init_black => 0.2,
  :albedo_white => 0.75,
  :albedo_black => 0.25,
  :surface_albedo => 0.4,
  :solar_change => 0.005,
  :solar_luminosity => 1.0,
  :scenario => :default,
  :seed => 165,
)
# Создаём список всех комбинаций
params_list = dict_list(param_dict)
```

```

for params in params_list
model = daisyworld(;params...)
daisycolor(a::Daisy) = a.breed
plotkwargs = (
agent_color=daisycolor,
agent_size = 20,
agent_marker = '□',
heatarray = :temperature,
heatkwargs = (colrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature,
plotkwargs...)
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature,
plotkwargs...)
plt1_name = savename("daisyworld",params) * "_step01" * ".png"
plt2_name = savename("daisyworld",params) * "_step04" * ".png"
plt3_name = savename("daisyworld",params) * "_step40" * ".png"
save(plotsdir(plt1_name), plt1)
save(plotsdir(plt2_name), plt2)
save(plotsdir(plt3_name), plt3)
end

```